Critical Path Analysis – Test Notes

Grade 9 EMS · Oct 29, 2026 · @LL

Key definitions

Learn these word for word. They are easy marks.

Term	Meaning
Critical path analysis (CPA)	Planning the sequence of activities in a project to find the most efficient and quickest way of completing it.
Critical path	The sequence of activities that must be completed on time to prevent any delay in the overall project.
Critical activity	Any activity on the critical path.
Non-critical activity	An activity that can be delayed without extending the completion time of the project.
Node	A circle representing a point in time: the start or completion of an activity.
Activity	A task that takes time, shown as a line (with an arrow) linking two nodes.
Duration	How long an activity takes (minutes, hours, days, weeks or months).
Prerequisite / predecessor	The activity (or activities) that must be finished before another can start.
EST (earliest start time)	The earliest an activity can start, which depends on the previous activities being finished.
LFT (latest finish time)	The latest the previous activity can finish without delaying the whole project.
Float time	How long a non-critical activity can be delayed without delaying the project.
Concurrent activities	Activities that run at the same time (in parallel).

How to draw a network diagram

Every node is a circle split into three parts.

- **Left half:** the node number.
- **Top right:** the EST.
- **Bottom right:** the LFT.

Each activity is a line with an arrow. Write the **letter above** the line and the **duration below** it.

1. Draw one start node and number it 1. All networks start and end at a single node.
2. Draw lines (left to right) from the start node for every activity with **no prerequisite**. Don't put nodes on their ends yet.
3. Write each activity's letter above its line and its duration below it.
4. Move to the next activity that has a prerequisite. Put a node at the end of the prerequisite's line, then draw the new activity from that node.
5. If an activity has **two or more prerequisites**, make all of their lines meet in the **same node**. That node is the moment they are all finished.
6. Keep going until every activity is drawn, numbering nodes as you add them.
7. Join all the loose line ends into one **final node**. That node is the end of the project.

Always draw in pencil and in landscape, so you can fix mistakes and fit everything in.

How to calculate EST, LFT, critical path and float

The short version: **forward pass, take the highest; backward pass, take the lowest.**

Step 1: EST (forward pass), working left to right

- The start node's EST is always **0**.
- EST of the next node = EST of this node + the activity's duration.
- If two or more lines go **into** a node, work out each one and take the **highest**. The next activity can't start until *all* of them are done.

Step 2: LFT (backward pass), working right to left

- In the final node, **LFT = EST**.
- LFT of the earlier node = LFT of the later node − the activity's duration.
- If two or more lines come **out of** a node, work out each one and take the **lowest**.
- The start node's LFT should come back to **0**. If it doesn't, check your sums.

Step 3: Critical path

- The critical path runs through every node where **EST = LFT**.
- Mark critical activity lines with **//**.
- The project's length is the EST in the final node.
- Write the answer as the letters in order plus the time, for example: A, C, E, F, G = 35 hours.

Step 4: Float

Total float = LFT − EST − Duration

- Use the **LFT of the node at the end** of the activity and the **EST of the node at the start** of it.
- Critical activities always have a float of **0**. If you get a negative float, you've made a mistake.

Advantages and disadvantages

CPA saves time and money when the estimates are good, but it's only as reliable as those estimates.

Real-life example: low-cost airlines used CPA to cut a plane's turnaround time (landing to take-off) to **25 minutes**. That's half the time of rivals like British Airways, so each plane can make more journeys. CPA is also used a lot in **construction**, where many activities can run at the same time.

Advantages

- **Saves time:** it shows the minimum time a project can take and which activities can run at the same time (concurrently).
- **Better use of resources:** workers can be moved from non-critical activities (which have float) onto critical ones to speed the project up.
- **Stock and cash:** the business knows exactly when resources are needed, so it can order stock just in time (JIT). It avoids holding too much stock or cash, which helps working capital.
- **Forces detailed planning:** this reduces the risk of unexpected delays.
- **Control and review:** progress can be checked against the plan so delays are spotted and fixed early.
- **Focus:** managers know which activities can't be delayed, so they watch those most closely.
- **Customers:** the business can give customers accurate completion dates.
- **Delays:** the effect of an unavoidable delay can be worked out by adjusting the network.

- **Better decisions:** it links departments such as marketing, production and finance, and it ties in with cash-flow forecasts and budgets.

Disadvantages

- **Relies on estimates:** if the durations are wrong, the whole plan breaks down. Big projects are often one-offs the firm has never done before.
- **Inflexible:** managers may treat the times and order as fixed and miss chances to speed things up.
- **One delay can stop everything:** activities are strictly timetabled, so if a critical activity runs late, hired equipment or workers may already be booked elsewhere (for example, a crane hired out to another company).
- **Complex projects** are hard to draw accurately; activities may need to be split into mini-projects.
- **Focuses on speed,** not quality or customer needs, so it doesn't always raise profit.
- **No guarantee of success:** resources may not be as flexible as managers hope when they try to use the float.

Exam tips and common mistakes

- **Highest going forward, lowest going back.** Mixing these up is the most common mistake.
- **Check your start node.** Its LFT must come back to 0. If it doesn't, you've made an error somewhere.
- **Lines with the same prerequisites meet in one node.** If F needs D and E, both lines end in the same circle.
- **An activity with no prerequisite** starts at node 1. If nothing depends on it, it runs straight to the final node.
- **Label everything:** node numbers, EST, LFT, letters above the lines, durations below, and // on the critical path. Marks are given for a "fully labelled" diagram.
- **Give units** in your answer, for example "50 weeks", not just "50".
- **"Will it finish on time?" questions:** compare the critical path length to the deadline, say yes or no, and state the difference (for example, "Yes, 2 weeks early").
- **Changed durations**: redo the whole forward and backward pass. The critical path can move to a different route.
- **Float formula:** LFT of the end node − EST of the start node − duration. Critical activities = 0.
- **Theory questions:** give the point *and* explain it. For example: "CPA relies on estimates. If an estimate is wrong, the whole plan is wrong, especially for one-off projects."